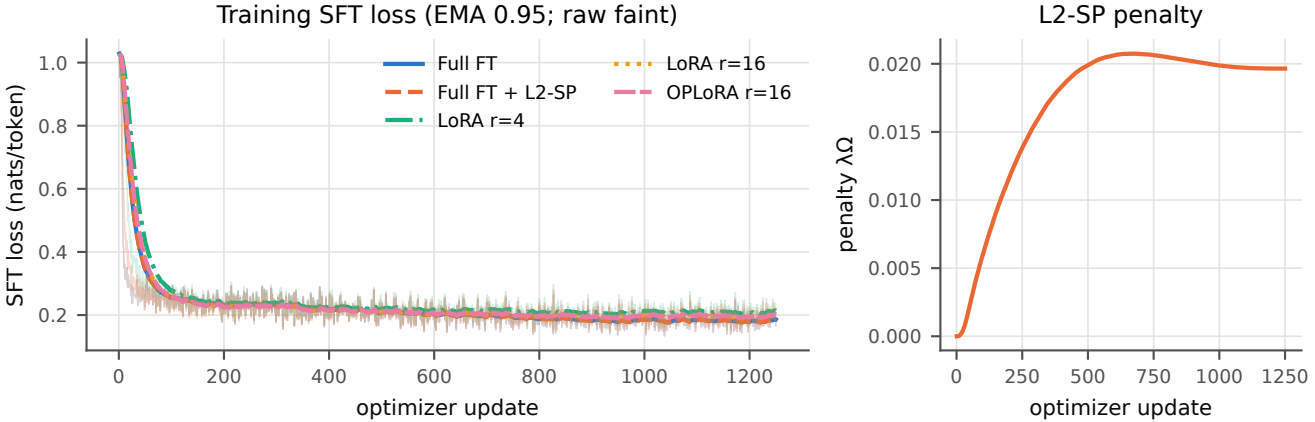

Training SFT loss (EMA 0.95; raw faint)
SFT loss (nats/token)
1.0
0.8
0.6
0.4
0.2
0
200
400
600
800
1000
1200
optimizer update
Full FT
Full FT + L2-SP
LoRA r=4
LoRA r=16
OPLoRA r=16
L2-SP penalty
penalty λΩ
0.020
0.015
0.010
0.005
0.000
0
250
500
750
1000
1250
optimizer update